

Roll No.

TCE-801

B. TECH. (CIVIL ENGINEERING) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

EARTHQUAKE RESISTANT DESIGN OF BUILDINGS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the various types of equations of motions considered in dynamic analysis. (CO1)

OR

- (b) Find the solution for SDOF for free undamped case. (CO1)
2. (a) A convenient way to determine damping in a system in a damp free case of vibration the dynamic properties of system : (CO1)

$W=50 \text{ N}$, $k = 55 \text{ N/cm}$, $C = 0.3\text{-sec/cm}$.

Find :

- (i) The natural frequency,
- (ii) Damping ratio,
- (iii) Logarithmic decrement, and
- (iv) Ratio of two successive amplitudes.

P. T. O.

(2)

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OR

(b) Write short notes on the following :

(CO1)

- (i) Angular frequency
- (ii) Damped frequency
- (iii) Damping ratio
- (iv) Logarithmic decrement

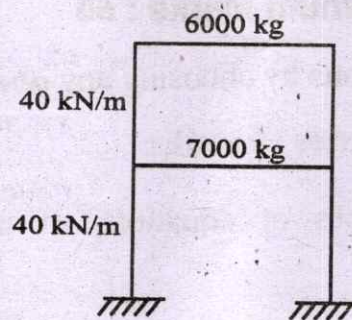
3. (a) Formulate an equation of motion for MDoF.

(CO2)

OR

(b) Draw the mode shapes for given problem :

(CO2)



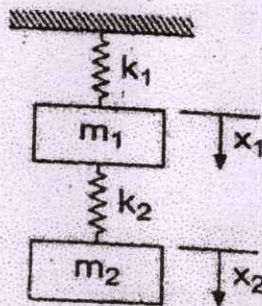
4. (a) Derive the ratio of two amplitudes of vibrations of the two masses under steady state conditions.

(CO2)

OR

(b) Obtain the frequency equation for the system as shown in Fig. below :

(CO2)



(3)

5. (a) What do you mean by damping ? Discuss the various types of damping. (CO2)

OR

- (b) Derive an expression of natural frequency of free longitudinal vibration of undamped free vibration by Spring Mass Energy System. (CO2)

1. When the following conditions are met, the following shall apply:

(a)

(b) When the following conditions are met, the following shall apply:

(c)

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TCE-813

**B. TECH. (CIVIL ENGINEERING)
(EIGHTH SEMESTER)**

MID SEMESTER EXAMINATION, March, 2024

HYDROPOWER ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any sub-questions among (a) and (b) in each main question.

(iii) Total marks for each main question are ten.

1. (a) Explain the necessity to predict future load demand. Describe the methods of load forecasting. (CO1)

OR

- (b) When a run-river plant operates as a peak load station with a weekly load factor of 20% all its capacity is firm capacity. What must be the minimum flow in the river so that the station may serve as the base load station ? It is given that rated installed capacity of generator = 10,000 kW, Operating head = 15 m, Plant efficiency = 80%. Estimate the daily load factor of the plant if the stream flow is 15 cumec ? (CO1)

P. T. O.

2. (a) A runoff river plant on a stream have a inflow of $20 \text{ m}^3/\text{s}$ and net head of 30 m with provision of pondage to meet daily peak demand with a load factor of 60%. Determine the power generation capacity of a plant at 80% of overall efficiency. The plant runs at a peak station for 3 hrs and balance period in a day for average load. Determine the amount of pondage is needed. (CO1)

OR

- (b) Describe in detail all the constraints in hydropower development. (CO1)
3. (a) The yearly output of a base load plant with 25 MW installed capacity is $125 \times 10^6 \text{ kWh}$. The plant takes a peak of 22.5 MW. Calculate the annual load factor and capacity factor. (CO1)

OR

- (b) A hydropower plant has an installed capacity of 30,000 kW. The yearly output of the plant is $150 \times 10^6 \text{ kWh}$. The peak load is 25,000 kW : Determine (a) annual load factor, (b) plant use factor, and (c) capacity factor. (CO1)
4. (a) In a hydroelectric station, water is available at the rate of $185 \text{ m}^3/\text{s}$ under a head of 15 m the turbine run at speed of 150 r.p.m with overall efficiency of 85%. Find the number of turbine required if they have the maximum specific speed of 440 ? (CO2)

(3)

OR

- (b) An isolated pumped storage power plant has a head race tunnel 4 m diameter and 700 m long. The gross head of water is 300 m and the power house discharges directly into the lower reservoir. The flow rate is 60 cumecs and friction factor is $f = 0.018$. If the overall efficiency of pumping and generation are 85% and 88% respectively, estimate the plant efficiency. (CO2)
5. (a) Calculate the specific speed of a turbine and suggest the type of turbine required for a river having a discharge of 240 L/s with an available head of 45 m. Take the efficiency of turbine = 82% and speed = 450 r.p.m. (CO2)

OR

- (b) Explain the Diversion Canal Plant and its different layouts with a neat sketch. (CO2)

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TCE-851

**B. TECH. (CE) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024**

DISASTER MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Classify natural disasters that your country deals with and enlist various such disasters that your region (State) has faced in past as per their year of occurrence. (CO1)

OR

- (b) Define the terms 'Hazards' and 'Vulnerability'. Write *five* possible causes of forest fires. (CO1)

2. (a) Can you make use of the fact that the man-made disasters have a long-term impact than a natural disaster on the environment ? Explain. (CO1)

OR

- (b) How would you differentiate sociological and technological disaster ? Explain with examples. (CO1)

P. T. O.

(2)

3. (a) How would you contrast the impact of disaster on victims and on the society (not on the nature) ? (CO1)

OR

- (b) Write a short note on indigenous practices that is done by the people in order to decrease the vulnerability with the help of suitable examples. (CO1)

4. (a) What facts would you suggest explaining positive realms of a disaster on the development ? (CO2)

OR

- (b) "Development can increase the vulnerability." Is the statement correct ? List down some points to support your answer. (CO2)

5. (a) How would you contrast the impact of climate change on disaster in terms of declining biodiversity, human migration and decrease in agricultural yield ? (CO2)

OR

- (c) What element would you choose to explain differential impacts of a disaster ? (CO1)